

DESIGN OF FERROELECTRIC FET-BASED SYMMETRIC COMPUTING-IN-MEMORY ARRAY WITH SIMULTANEOUS WEIGHT GRADIENT CALCULATION AND WEIGHT UPDATE FOR ON-CHIP LEARNING

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ABSTRACT

In this work, a novel ferroelectric FET (FeFET) based symmetric computing-in-memory (CIM) array enabling simultaneous weight gradient (ΔW_n) calculation and weight update is proposed for on-chip learning with reduced latency. By utilizing stochastic computing (SC) scheme with novel bipolar pulse streams for ΔW_n calculation and utilizing the output pulses of SC for update of FeFET weights, the ΔW_n calculation and weight update of backpropagation (BP) learning are in-situ realized simultaneously for the whole array of CIM cell with special designed tri-state XNOR gate. Moreover, through the design of symmetric differential input and readout based on reference FeFET, four-quadrant vector-matrix multiplication (VMM) of configurable forward and backward propagation are implemented in the FeFET-based CIM array. Based on the design, image recognition's neural network is demonstrated with a high accuracy of 95.33% and reduced hardware cost, showing great potential for on-chip learning.

INTRODUCTION

Non-volatile memory (NVM) based computing-in-memory (CIM) architecture has been proven to have the advantages of high energy efficiency and density [1]. It has been widely studied for both inference and training processes of deep neural networks (DNNs), including forward and backward propagation (Fig. 1a). Among NVMs, hafnium-based ferroelectric (FE) FETs (FeFETs) are highly regarded for their high CMOS compatibility, low energy consumption and high on/off ratio [2, 3]. Recently, in order to enhance on-chip learning performance, the FeFET weight cells with specially designed gate stack structures for broadened distribution of coercive field (E_c) have been proposed to improve the linearity and symmetry during weight update [4, 5]. A symmetric CIM with a cell structure of 1FeFET+2T [6] was further proposed to support forward and backward propagation. However, the existing works still require extra peripheral circuits for weight gradient calculation and row-wise writing of weight update [7], suffering from high hardware cost and latency.

In this work, a novel FeFET-based symmetric CIM array is proposed for on-chip learning, which enables

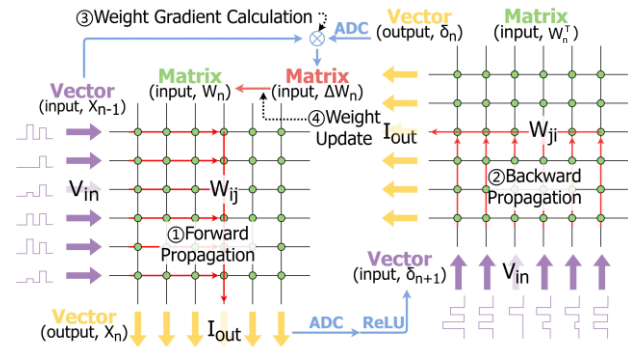


Figure 1: The on-chip implementation of a DNN network based on the BP learning algorithm in a CIM array.

simultaneous weight gradient calculation and weight update of the whole array within a single computing cycle through a novel bipolar pulse-modulated stochastic computing (SC) design. Moreover, configurable forward and backward propagations are realized with a four-quadrant range through the array design with symmetric differential input and readout. Based on this design, the learning process of image recognition is demonstrated with high accuracy and low latency.

FEFET-BASED SYMMETRIC CIM ARRAY WITH SIMULTANEOUS WEIGHT GRADIENT CALCULATION AND WEIGHT UPDATE

The architecture of the proposed FeFET-based CIM system for on-chip learning is shown in Fig. 2a. It consists of a symmetric FeFET-CIM array composed of stochastic computing (SC) based weight update (SWU-SCIM) cell, along with peripheral circuits for configurable operation modes of weight update and forward/backward propagation. Each SWU-SCIM cell contains 2 FeFETs (T1 and T2) and 4 transistors (M1~M4) (Fig. 2b). T1 is a weight device with variable channel conductance, while T2 is a reference weight device with fixed conductance, and both are used for four-quadrant vector-matrix multiplication (VMM). M1~M4 forms a tri-state output XNOR (TS-XNOR) for bipolar SC based weight gradient calculation and weight update. The stochastic translators (STRs) are used to convert inputs into random bitstreams

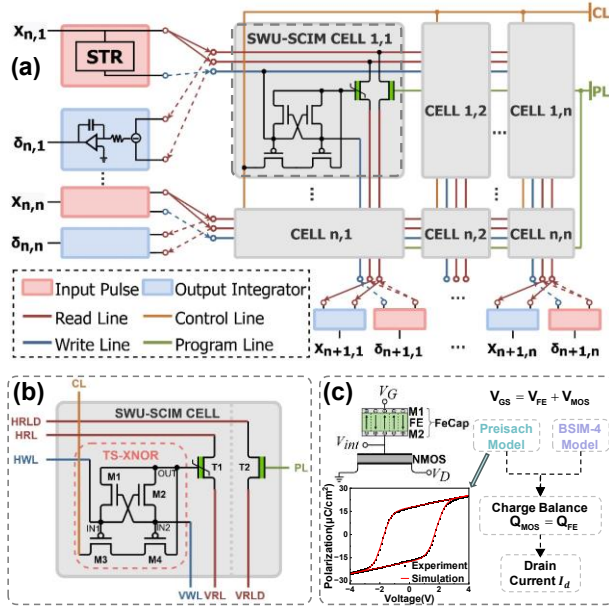


Figure 2: (a) Circuit diagram of a FeFET based SWU-SCIM cell. The read ports are horizontal read line (HRL), differential HRL (HRLD), vertical read line (VRL), and differential VRL (VRLD). The write ports are horizontal/vertical write line (HWL/VWL). The control ports are program line (PL) and control line (CL). (b) Circuit diagram of a symmetric FeFET-based CIM array with simultaneous weight gradient calculation and weight update. (c) The modelling framework of FeFET.

in the weight gradient calculation and weight update mode. Alternating subtractors and integrators in the row and column directions are utilized for configurable forward and backward calculation.

To demonstrate and analyze the weight update behavior, the FeFET device model is developed based on the BSIM-4 MOSFET model and the multi-domain Preisach FE model [8]. Simulations performed in HSPICE show good agreement with the experimental data (Fig. 2c).

SYMMETRIC WEIGHT GRADIENT CALCULATION AND WEIGHT UPDATE

In the backpropagation (BP) learning algorithm, activation vector x_{n-1} at layer $n-1$ and loss vector δ_n at layer n are inputs. ΔW_n at layer n is generated through vector-vector multiplication (VVM) of x_{n-1} and δ_n (Fig. 1a). ΔW_n is added to W_n at layer n to complete the weight update. In this work, the stochastic computing (SC) scheme [9, 10], which uses pulse-modulated probabilistic bitstreams to simplify arithmetic operations with basic logic gates, is applied to simplify weight gradient calculation. Simultaneously, the pulse-width modulated outputs of SC can be directly applied to the FeFET gate for in-situ weight update.

For weight gradient calculation and weight update

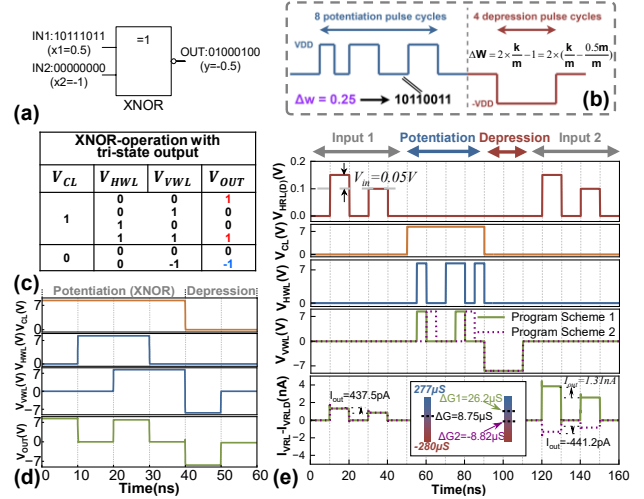


Figure 3: (a) Bipolar stochastic computing based on XNOR. (b) A bipolar pulse weight update scheme based on stochastic computing principles. (c) Truth table and simulated waveform diagram of TS-XNOR gate. (d) Simulated waveform diagram of the SWU-SCIM cell in the weight gradient calculation and weight update mode.

mode, the FeFET-CIM array is configured as indicated by the blue solid line in Fig. 2a. The array's HWL and VWL write ports are connected to the STRs modules on rows and columns. All read ports are grounded. The vector x_{n-1} and δ_n are converted to bipolar stochastic bitstreams by the STRs and then applied to HWLs and VWLs. The CL is set to V_{DD} for the potentiation pulse and to $-V_{DD}$ for the depression pulse.

For weight gradient calculation, XNOR logic is used for SC based multiplication of bipolar data (Fig. 3a). For weight update, bipolar pulse is designed to express signed ΔW_n (Fig. 3b). Here, k is the number of logical "1" in the bitstreams, m is the bitstream length. Through the combined processes of potentiation and depression, the pulse-width modulated outputs proportional to ΔW_n are generated. In potentiation programming, the XNOR logic output is directly applied to the gate of FeFET with k cycles of V_{DD} . In depression programming, $0.5m$ cycles of $-V_{DD}$ are applied. A TS-XNOR (Fig. 2b) is proposed for bipolar pulse, which has three inputs: gate control CL, symmetric logic inputs IN1 (V_{HWL}) and IN2 (V_{VWL}), and output OUT (V_{OUT}). Its logic operation and functionality are shown in Fig. 3c and Fig. 3d. During HSPICE simulation, in two cases with different orthogonal vector inputs (Fig. 3e), the change in stored conductance was proportional to the computed weight gradient. So, the proposed bipolar SC based CIM array can complete symmetric weight gradient calculation and weight update for DNN on-chip learning.

CONFIGURABLE FOUR-QUADRANT FORWARD AND BACKWARD PROPAGATION

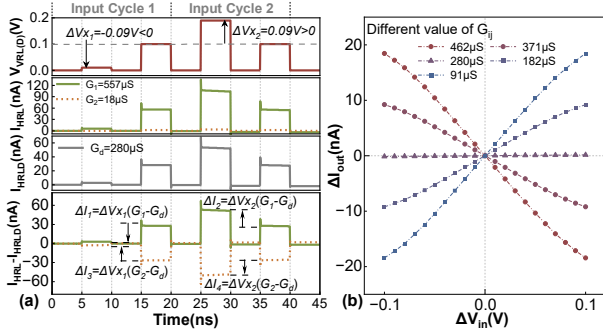


Figure 4: (a) Simulated waveform of the SWU-SCIM cell in forward propagation mode. (b) Simulated scatter plot of four-quadrant multiplication in the SWU-SCIM cell.

For forward propagation, x_n is generated through VMM between activation vector x_{n-1} and weight matrix W_n . For backward propagation, δ_{n-1} is generated through VMM between loss vector δ_n and weight matrix W_n^T . The signed VMM calculation is realized by using differential input and readout. The forward propagation mode is configured as indicated by the red solid lines in Fig. 2a. The row read ports (HRL, HRLD) are directly connected to the input data lines, and the column read ports (VRL, VRLD) are connected to the column subtractors and integrators, with all other ports grounded. The backward propagation mode is configured as indicated by the red solid lines in Fig. 2a, symmetrically mirroring the forward propagation mode.

For differential input, the differential encoded voltage V_x and $V_{max}/2$ applied over two cycles, and the outputs from the two cycles are subtracted. So, the equivalent input voltage is given by $V_{in} = V_x - V_{max}/2$. For differential storage and readout, T1 and T2 are programmed to G_{ij} and $G_{max}/2$, and the outputs from the two FeFETs are subtracted. The simulation results (Fig. 4ab) show that the differential input voltages can perform four-quadrant multiplication with the differential storage values and outputting differential currents, demonstrating that the proposed CIM array enables forward and backward propagation for DNN on-chip learning.

FEFET-BASED SCU-SCIM FOR ON-CHIP LEARNING OF DNN

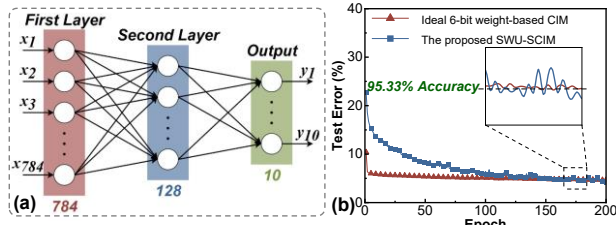


Figure 5: (a) Diagram of two-layer neural network architecture for MNIST image classification. (b) Simulated on-chip learning accuracy based on the proposed symmetric CIM array.

TABLE I. Comparison of CIM Designs for On-Chip Learning

	Nature [11]	IEDM [7]	IEDM [4]	JXCDC [12]	This work
Weight Type	1T-1R	21T-1C	1T-1FeFET	5FeFET	4T-2FeFET
Weight Precision	Analog	9	7	1	6
Forward Propagation	✓	✓	✓	✓	✓
Backward Propagation	×	✓	×	✓	✓
Error Calculation	✓	×	×	✓	✓
Weight Update	Only Last Layer	✓	✓	✓	✓
Synchronous Update	Only Last Layer	✓	×	×	✓
Learning Accuracy	96.20%	92.70%	94.10%	-	95.33%

Based on the proposed symmetric CIM array with simultaneous weight gradient calculation and weight update, the on-chip learning of a 2-layer neural network is demonstrated. The network architecture is shown in Fig. 5a, and the MNIST dataset is used for image recognition tasks. It achieves 95.33% accuracy after 200 iterations (Fig. 5b). Thus, the proposed CIM array and framework ensure reliable on-chip learning. Compared with other recent CIM works for on-chip learning (Table I), the proposed design supports both forward and backward propagations while enabling simultaneous weight gradient calculation and weight update for the whole array.

CONCLUSION

This work reports a novel FeFET-based symmetric CIM array for on-chip learning, enabling simultaneous weight gradient calculation and weight update through bipolar pulse based stochastic computing design. Moreover, configurable forward and backward propagations with a four-quadrant range are achieved through a symmetrical array design with differential input and readout. Based on the design, on-chip DNN learning for image recognition is demonstrated with high accuracy, highlighting its strong potential for edge AI.

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